AWS VPC Peering Project Report

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1. Project overview

I built two private networks (VPCs) on AWS and connected them with VPC peering. Servers in the two networks can now talk to each other using private IP addresses only, so no traffic goes over the public internet.

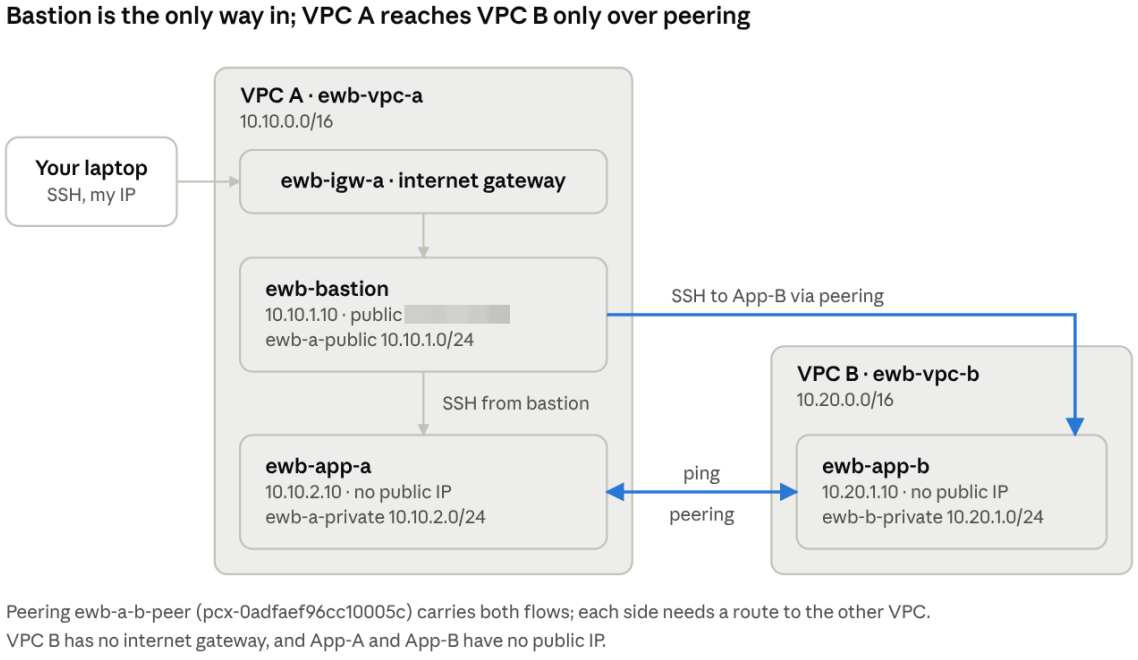
A peering link alone is not enough. Both sides also need a route to the other network, and the security group on the destination must allow the traffic.

Item	Detail
Student	Roshan
Project	AWS VPC Peering (EWB Courses student project)
Region	Asia Pacific (Mumbai), ap-south-1, zone ap-south-1a
Date	23 October 2026
Tags on every resource	Project = EWB-VPC-Peering, Student = Roshan

Services used: Amazon VPC (networks, subnets, route tables, internet gateway, peering, security groups), Amazon EC2 (three t3.micro servers running Amazon Linux 2023) and Amazon EBS (server disks). I did not use a NAT gateway.

2. Architecture

VPC A has a public server (the bastion) and a private server (App-A). VPC B has one private server (App-B) and no internet gateway. The two network ranges, 10.10.0.0/16 and 10.20.0.0/16, do not overlap, which peering requires.



Architecture diagram. I reach the bastion over SSH. From there, SSH and ping to VPC B go only through the peering connection.

Resource	Name	Range / IP	ID
VPC A	ewb-vpc-a	10.10.0.0/16	vpc-019d3611ee96f9bdf
VPC B	ewb-vpc-b	10.20.0.0/16	vpc-0c0a55a88f4bbb142
Public subnet	ewb-a-public	10.10.1.0/24	subnet-073d83e53439aa6d1
Private subnet A	ewb-a-private	10.10.2.0/24	subnet-063c720e55d1e3fa7
Private subnet B	ewb-b-private	10.20.1.0/24	subnet-093af10c119aabbb5
Internet gateway	ewb-igw-a	attached to VPC A	igw-0b41535d80c529a5f
Route table	ewb-rt-a-public	for ewb-a-public	rtb-0e2e6d38e6aa56828
Route table	ewb-rt-a-private	for ewb-a-private	rtb-0c04256f29641595d
Route table	ewb-rt-b-private	for ewb-b-private	rtb-01490f123ed02c97a
Security group	ewb-sg-bastion	VPC A	sg-074a2654d42d5a188
Security group	ewb-sg-app-a	VPC A	sg-023996b942743442a
Security group	ewb-sg-app-b	VPC B	sg-0cf7bf442c783f20a
Server	ewb-bastion	10.10.1.10 (has a public IP)	i-... (to add)
Server	ewb-app-a	10.10.2.10	i-07fb04299a6ed2097
Server	ewb-app-b	10.20.1.10	i-0fc3263e2f8e469ce
Peering connection	ewb-a-b-peer	VPC A to VPC B	pcx-0adfaef96cc10005c

3. Subnets, internet gateway and route tables

A subnet is public only when its route table sends traffic to an internet gateway. ewb-a-public does this, and the bastion in it has a public IP. The other two subnets have no internet route, and App-A and App-B have no public IP. Each subnet is linked to its own route table.

Final routes in the three route tables:

Route table	Subnet	Routes
ewb-rt-a-public	ewb-a-public	10.10.0.0/16 local; 0.0.0.0/0 to the internet gateway; 10.20.0.0/16 to the peering connection
ewb-rt-a-private	ewb-a-private	10.10.0.0/16 local; 10.20.0.0/16 to the peering connection
ewb-rt-b-private	ewb-b-private	10.20.0.0/16 local; 10.10.0.0/16 to the peering connection

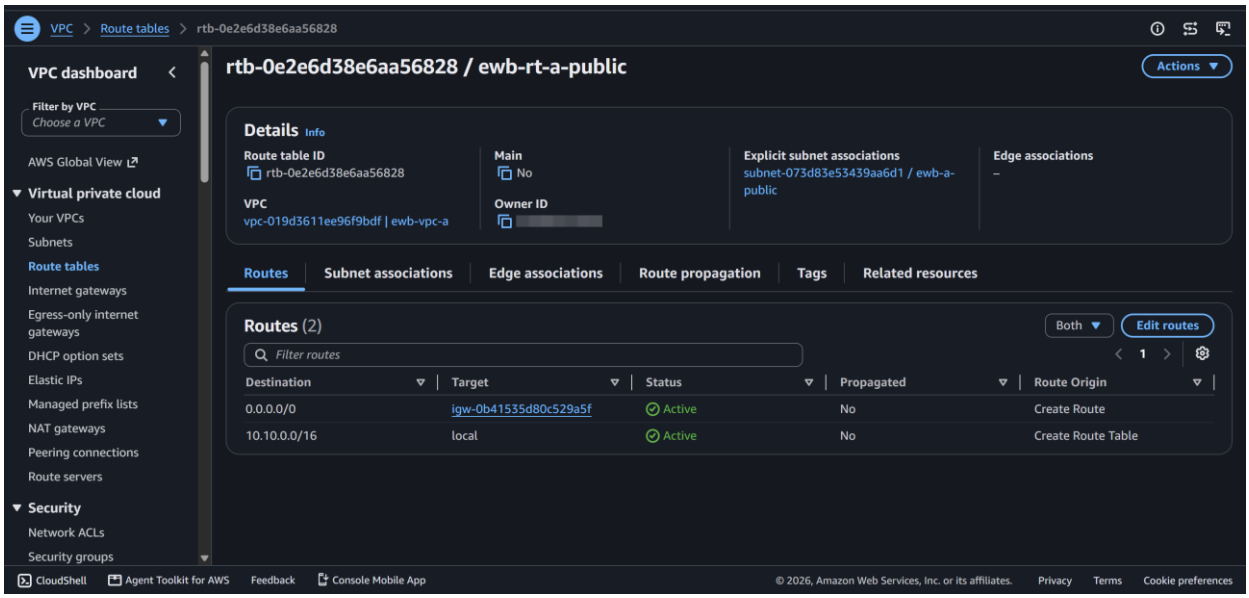


Figure 1. ewb-rt-a-public before peering. It has the local route and a route to the internet gateway, which makes its subnet public.

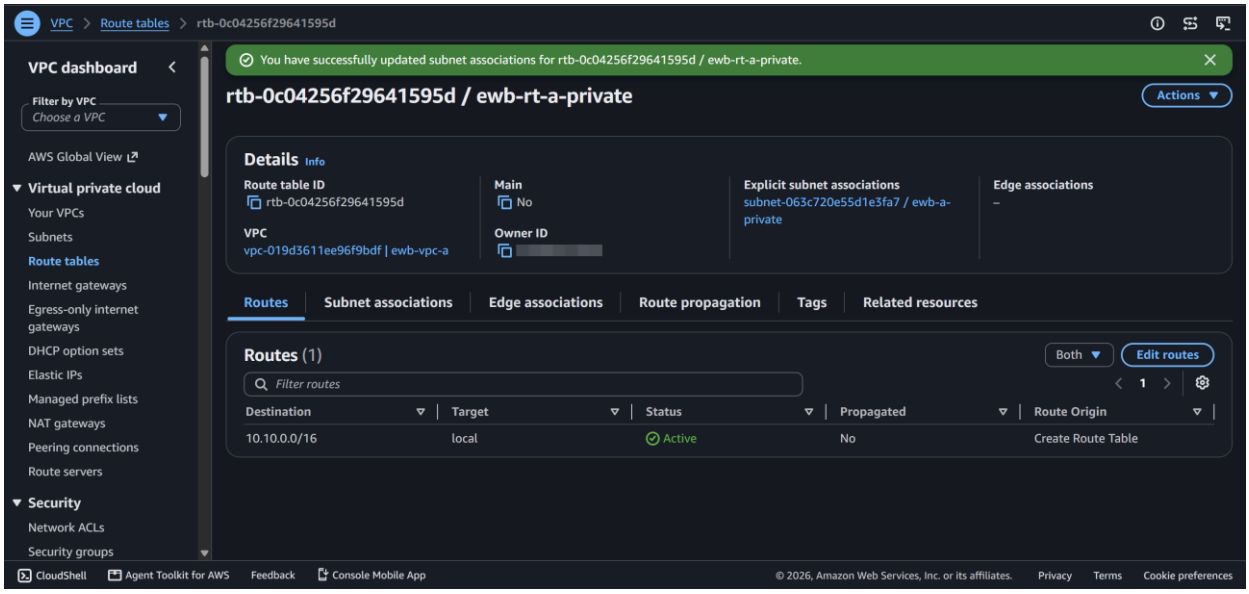


Figure 2. ewb-rt-a-private before peering. It has only the local route.

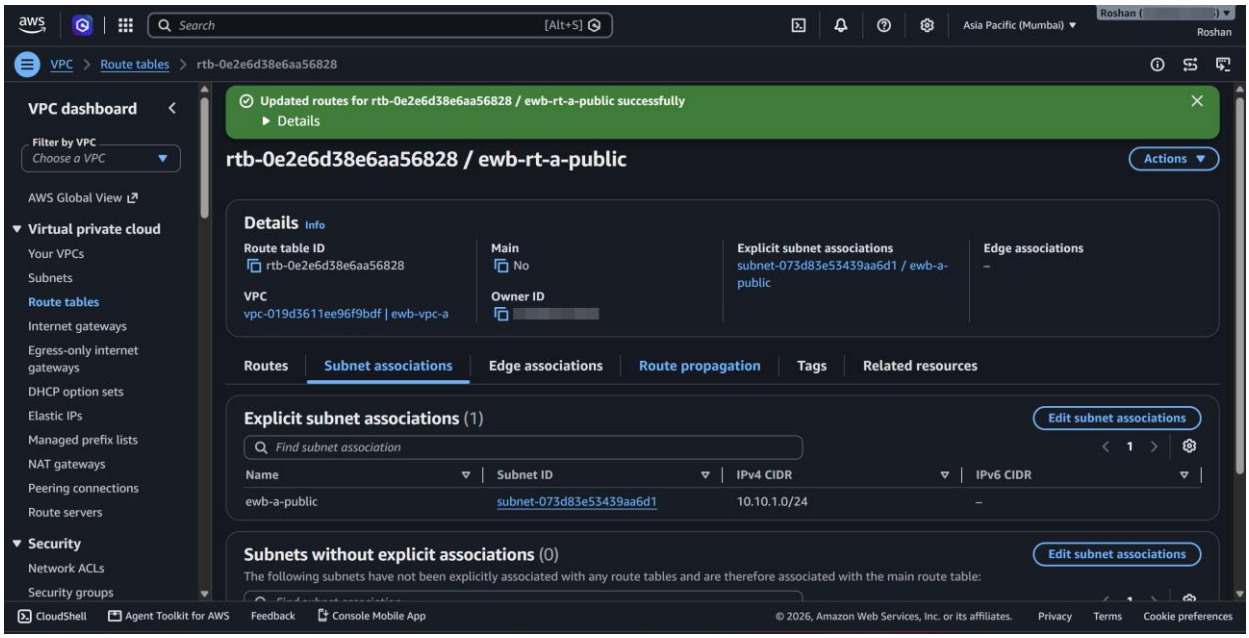


Figure 3. ewb-rt-a-public is linked to the subnet ewb-a-public (10.10.1.0/24).

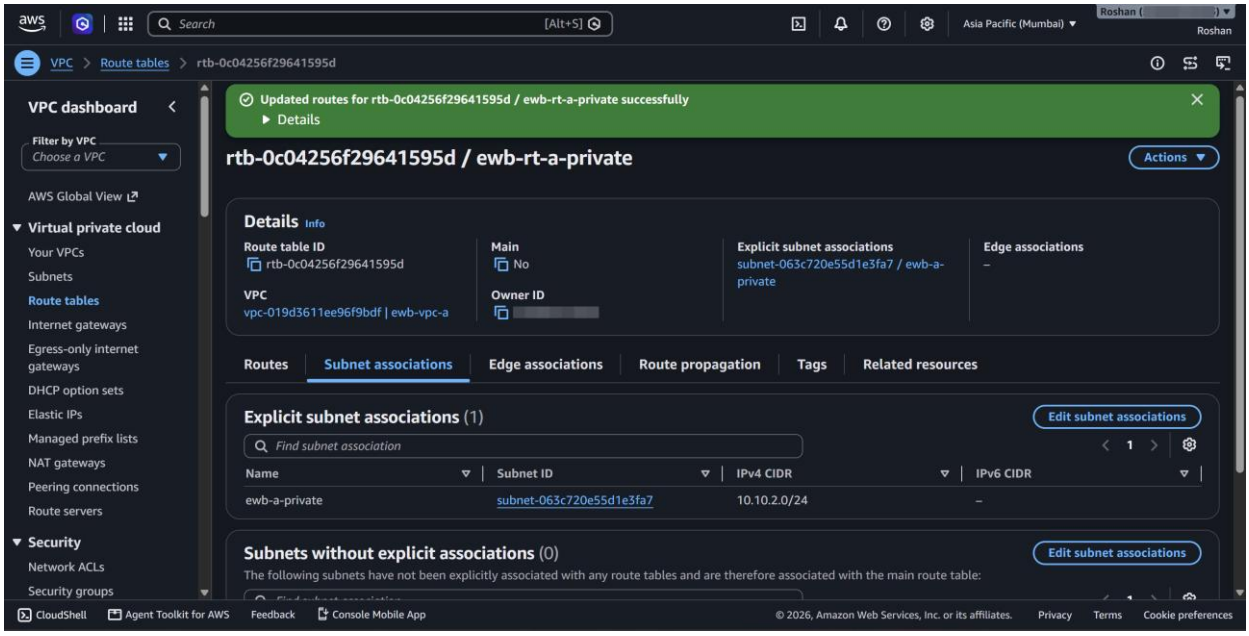


Figure 4. ewb-rt-a-private is linked to the subnet ewb-a-private (10.10.2.0/24).

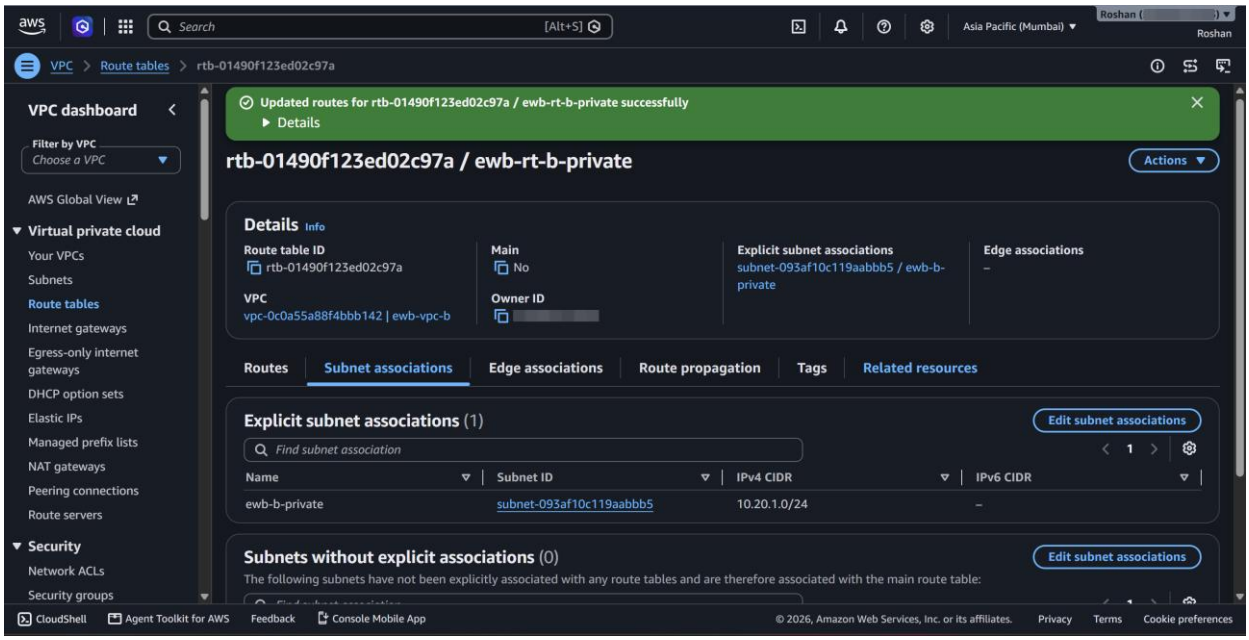


Figure 5. ewb-rt-b-private is linked to the subnet ewb-b-private (10.20.1.0/24).

4. Peering connection and routes

I sent a peering request from VPC A (requester) to VPC B (accepter) and accepted it. Active only means the link exists. Traffic flows only when both sides have a route to the other network and the destination's security group allows it.

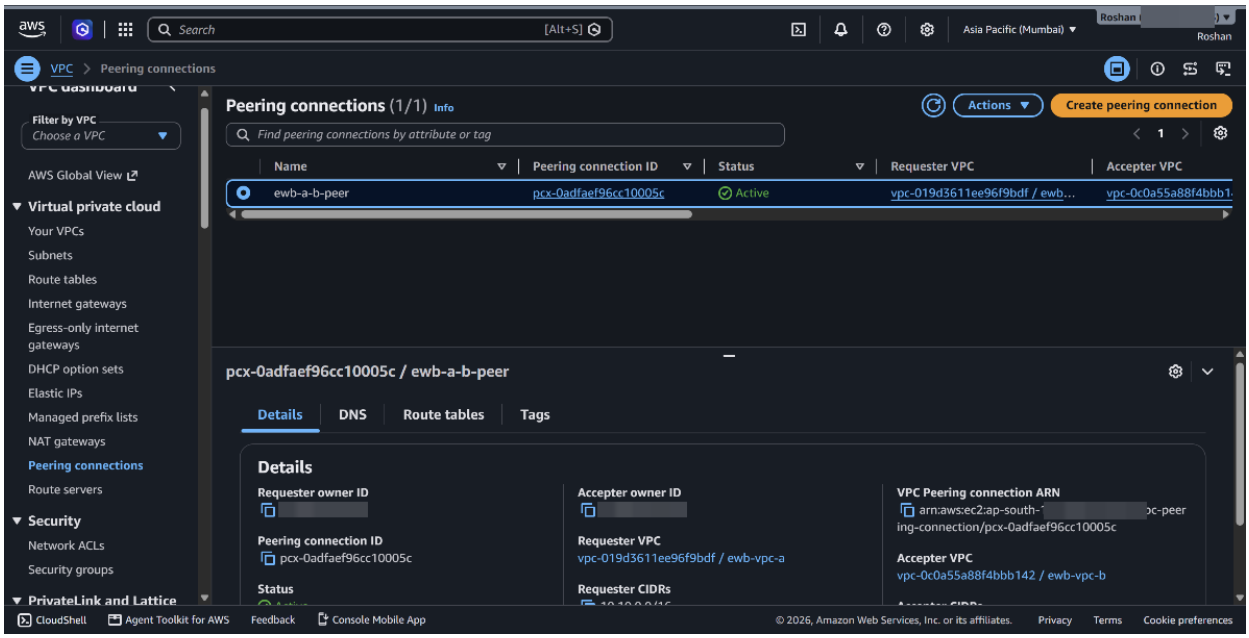


Figure 6. The peering connection ewb-a-b-peer is Active. It connects ewb-vpc-a and ewb-vpc-b.

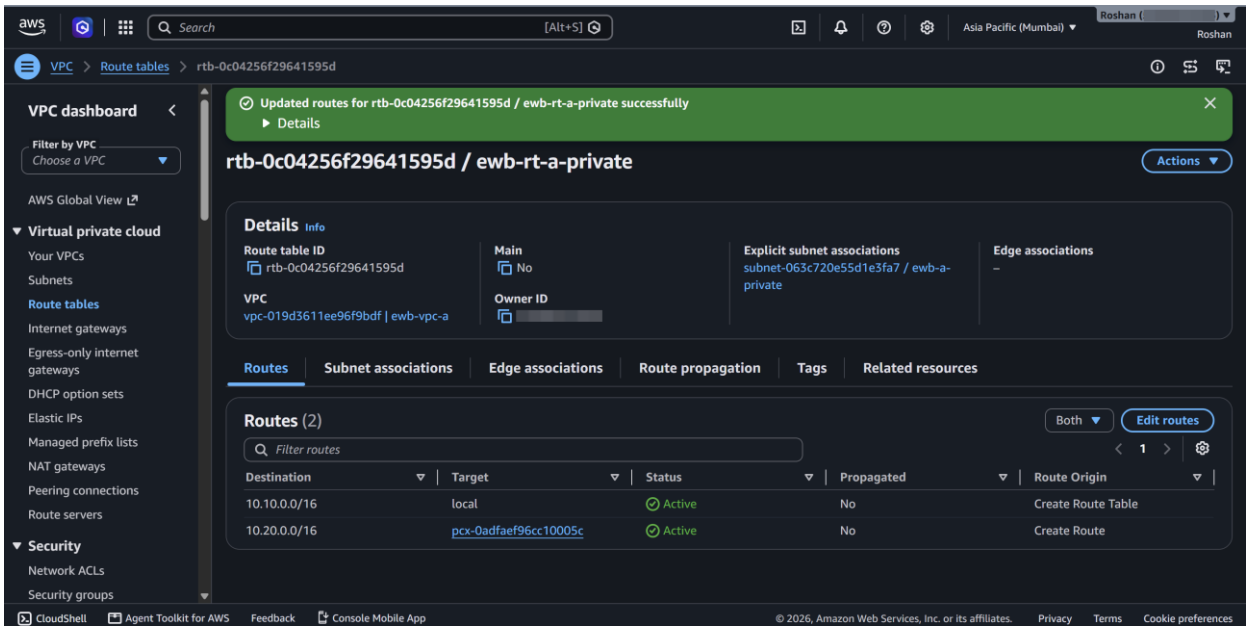


Figure 7. ewb-rt-a-private now sends 10.20.0.0/16 to the peering connection. This is the forward route from App-A to VPC B.

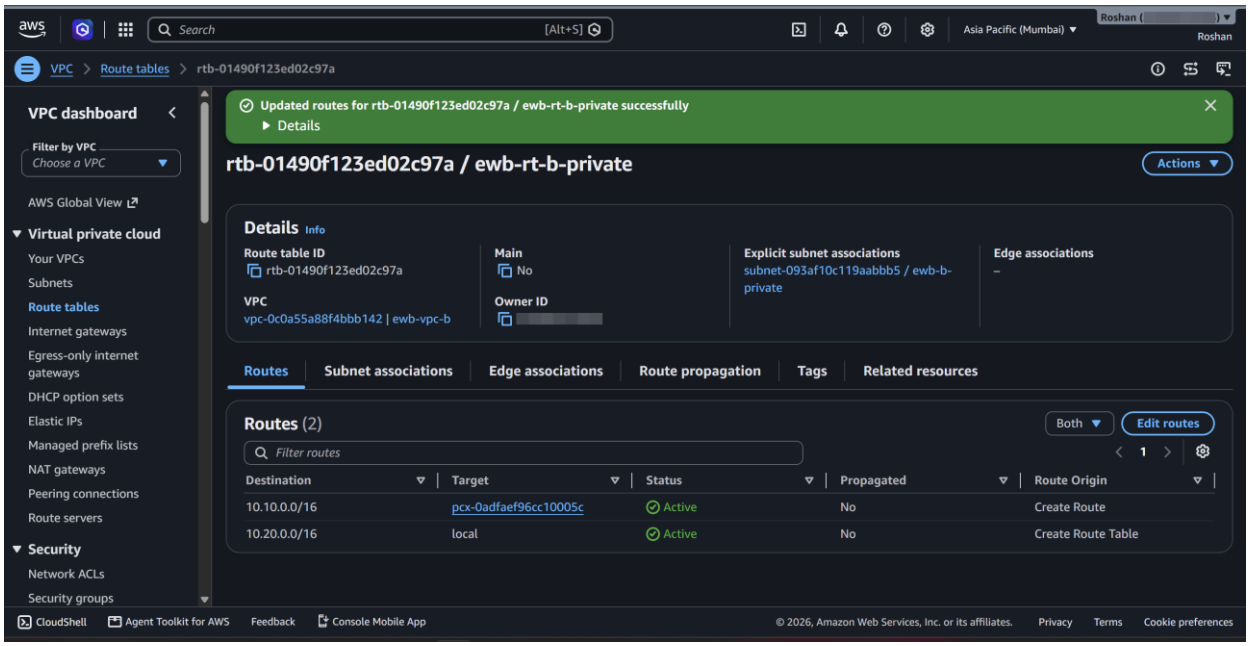


Figure 8. ewb-rt-b-private now sends 10.10.0.0/16 to the peering connection. This is the return route back to VPC A.

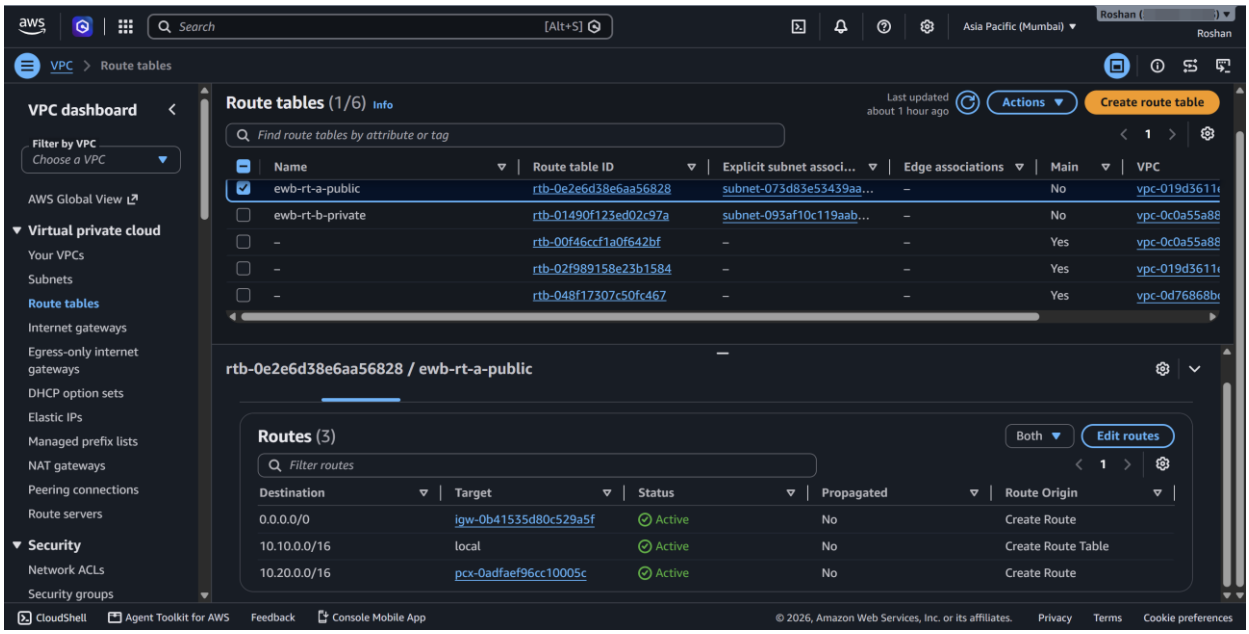


Figure 9. ewb-rt-a-public now has three routes: local, the internet gateway and the peering connection. The bastion needs the peering route to reach App-B over SSH.

5. Security groups and SSH

Each security group allows only what its server needs. I wrote inbound rules only; outbound stays on the default (allow all).

Security group	Used by	Inbound rules
ewb-sg-bastion	ewb-bastion	SSH (port 22) from my public IP only
ewb-sg-app-a	ewb-app-a	SSH from 10.10.1.10/32 (bastion); ping (ICMP Echo Request) from 10.20.1.10/32 (App-B)
ewb-sg-app-b	ewb-app-b	SSH from 10.10.1.10/32 (bastion); ping (ICMP Echo Request) from 10.10.2.10/32 (App-A)

I use SSH through the bastion (ProxyJump), so my private key stays on my laptop and is never copied to a server. App-B sees the bastion's private IP, 10.10.1.10, as the SSH source because the bastion makes the second connection from inside VPC A.

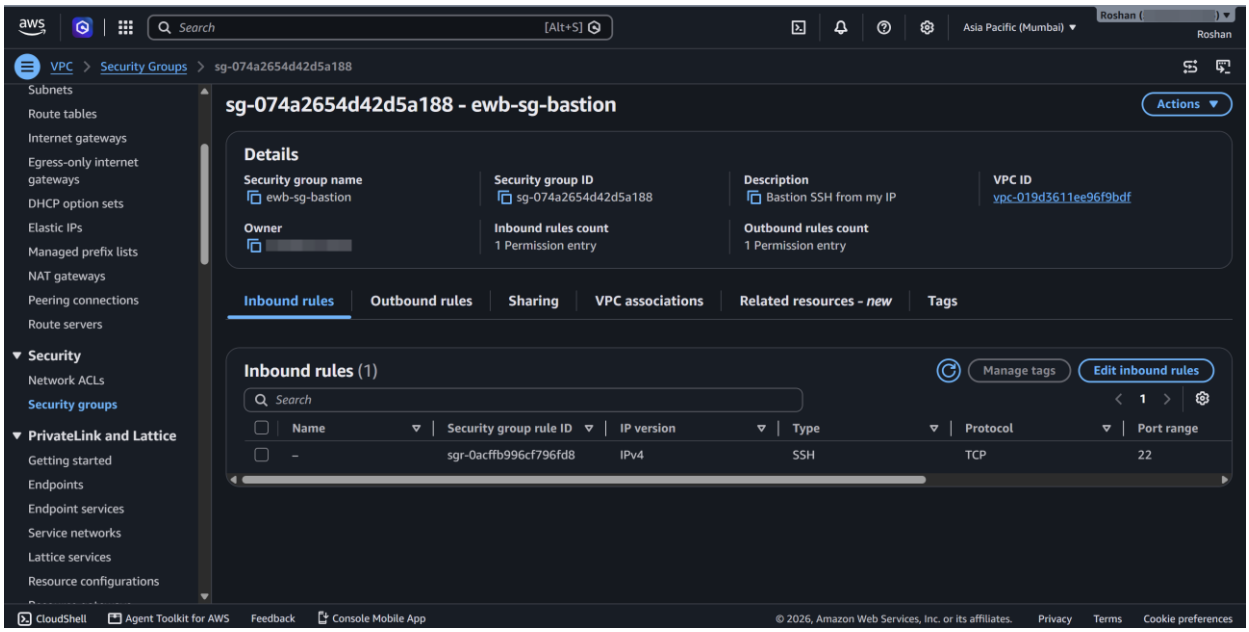


Figure 10. ewb-sg-bastion has one inbound rule: SSH on port 22.

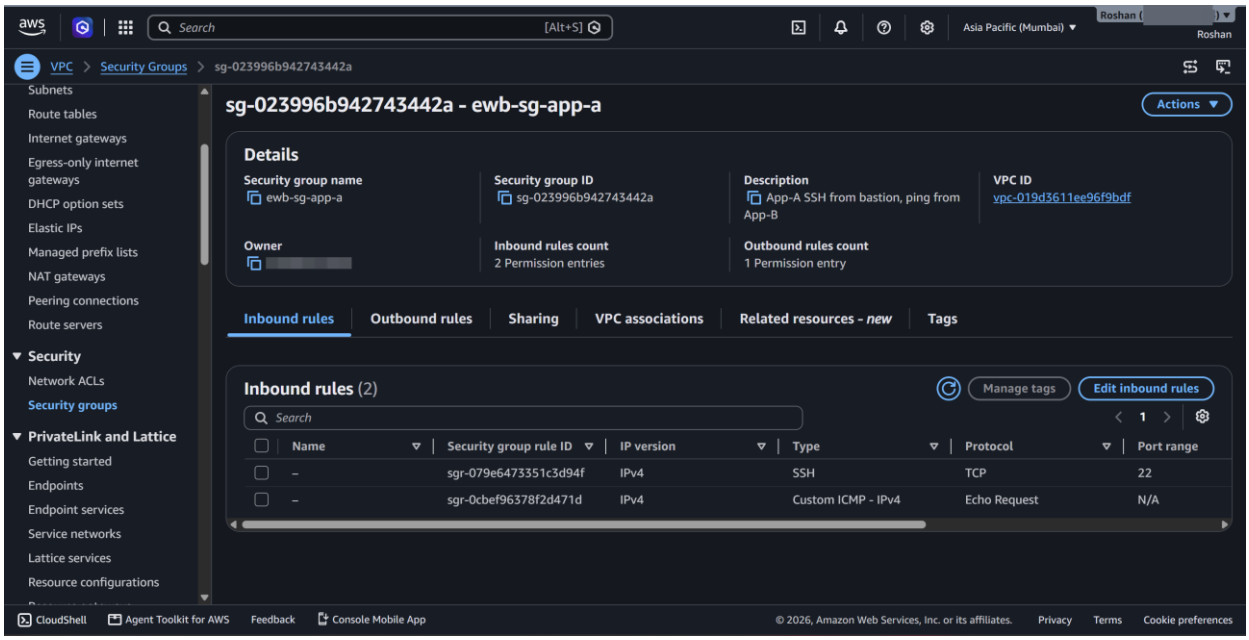


Figure 11. ewb-sg-app-a has two inbound rules: SSH on port 22 and ICMP Echo Request (ping).

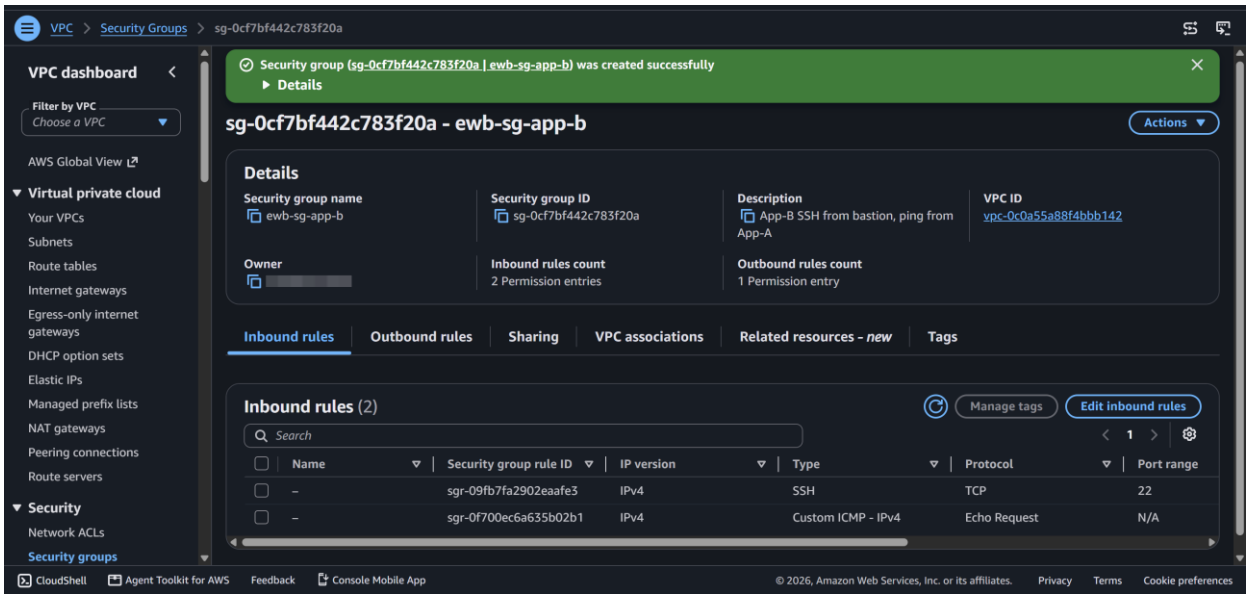


Figure 12. ewb-sg-app-b has two inbound rules: SSH on port 22 and ICMP Echo Request (ping).

6. Test results

Before peering and routes, the ping failed. After them, it worked in both directions with 0% packet loss.

Test	What I did	Result	Figure
01 Before peering	Ping from App-A to 10.20.1.10	100% packet loss	13
02 Forward path	Ping from App-A to App-B	4 of 4 replies, 0% loss	14
03 Reverse path	Ping from App-B to App-A	4 of 4 replies, 0% loss	15
04 Private SSH	Laptop to bastion to App-B	Login worked	16
05 No public IP	Check App-A and App-B details	Public IPv4 is empty on both	17, 18
06 Controlled failure	Ping App-A to App-B with the ping rule removed	To add	19
07 Recovery	Same ping after adding the rule back	4 of 4 replies, 0% loss	20, 21

Test 01: before peering

```
PS C:\Windows\System32> ssh ewb-bastion
The authenticity of host '[redacted] ([redacted])' can't be established.
ED25519 key fingerprint is SHA256:4sCv6SJNtXVA/qRmvRKo8xcoLKE4m5EOsZx+Y7XebRU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[redacted]' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-1-10 ~]$ hostname -I
10.10.1.10
[ec2-user@ip-10-10-1-10 ~]$ exit
logout
Connection to [redacted] closed.
PS C:\Windows\System32> ssh ewb-app-a
The authenticity of host '10.10.2.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:0t8scdbKmazFQZFkrzANZiHV6SWHXbZUiUpWX+RnnA4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.2.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-10-2-10 ~]$ hostname -I
10.10.2.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3126ms

[ec2-user@ip-10-10-2-10 ~]$
```

Figure 13. I log in to the bastion (10.10.1.10) and App-A (10.10.2.10), then ping 10.20.1.10. All 4 packets are lost, as expected, because there is no peering and no route yet.

Test 02: forward path (App-A to App-B)

```
PS C:\Windows\System32> ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 23 07:08:51 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ hostname -I
ping -c 4 10.20.1.10
10.10.2.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.811 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.521 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.495 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.523 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3117ms
rtt min/avg/max/mdev = 0.495/0.587/0.811/0.129 ms
[ec2-user@ip-10-10-2-10 ~]$ exit
logout
Connection to 10.10.2.10 closed.
```

Figure 14. On App-A, ping 10.20.1.10 gets 4 replies and 0% packet loss.

Test 03: reverse path (App-B to App-A)

```
[ec2-user@ip-10-20-1-10 ~]$ hostname -I
10.20.1.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.497 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.523 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.521 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.503 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3145ms
rtt min/avg/max/mdev = 0.497/0.511/0.523/0.011 ms
[ec2-user@ip-10-20-1-10 ~]$ exit
logout
Connection to 10.20.1.10 closed.
```

Figure 15. On App-B, ping 10.10.2.10 gets 4 replies and 0% packet loss.

Test 04: private SSH to App-B through the bastion

```
PS C:\Windows\System32> ssh ewb-app-b
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:A33IVx0midsPqRzfQWkM/ZEoRCM7w2bSv8KEsTsBuwA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$ hostname -I
10.20.1.10
```

Figure 16. I log in to App-B through the bastion. The prompt shows ip-10-20-1-10 and hostname -I shows 10.20.1.10.

Test 05: no public IP on App-A and App-B

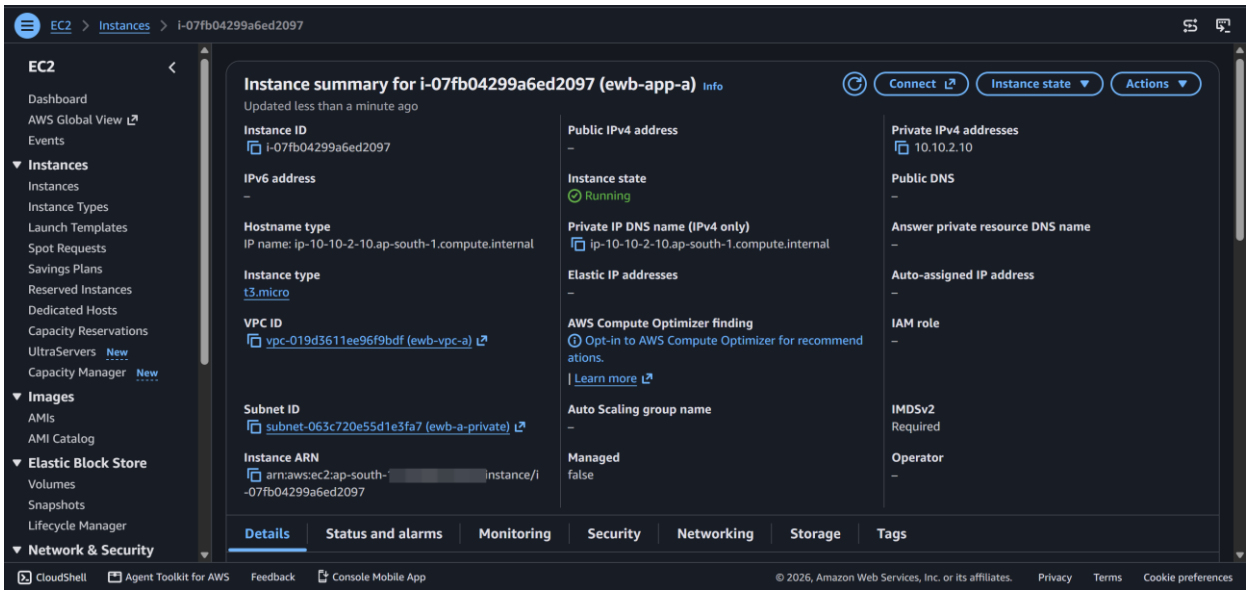


Figure 17. ewb-app-a has no public IPv4 address. Its private IP is 10.10.2.10.

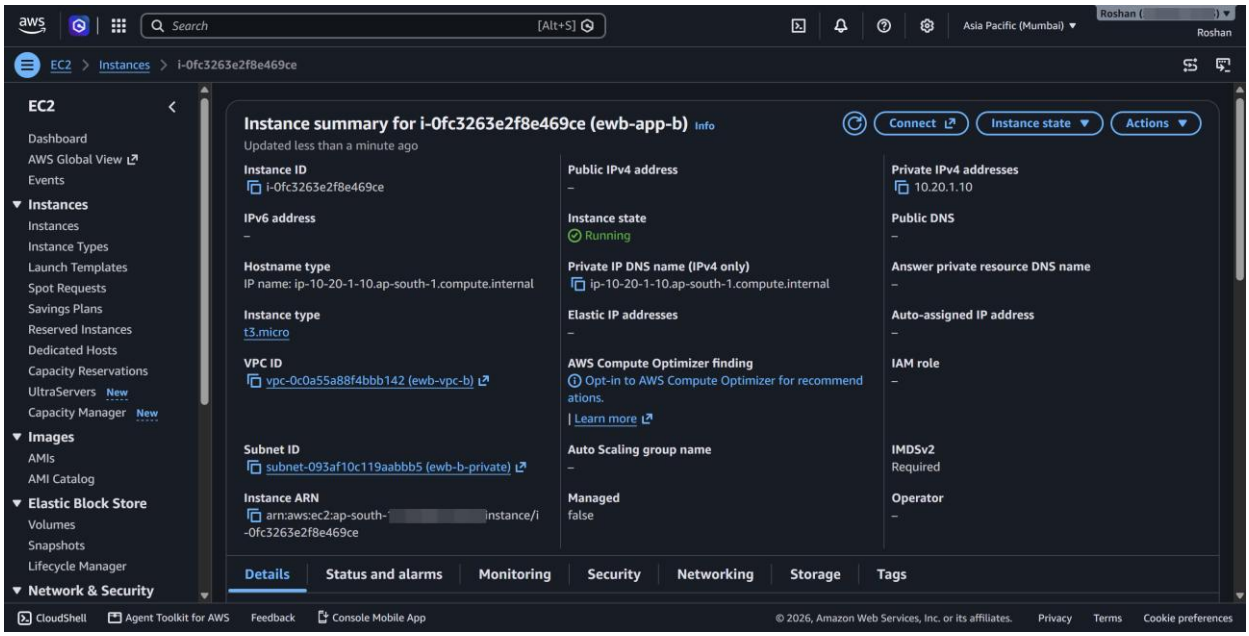


Figure 18. ewb-app-b has no public IPv4 address. Its private IP is 10.20.1.10.

Tests 06 and 07: controlled failure and recovery

```
PS C:\Windows\system32> ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 23 10:27:25 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.507 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.514 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.509 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.508 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3108ms
rtt min/avg/max/mdev = 0.507/0.509/0.514/0.002 ms
[ec2-user@ip-10-10-2-10 ~]$
```

Figure 19. After I delete the Echo Request rule, ewb-sg-app-b has only one inbound rule left (SSH on port 22).

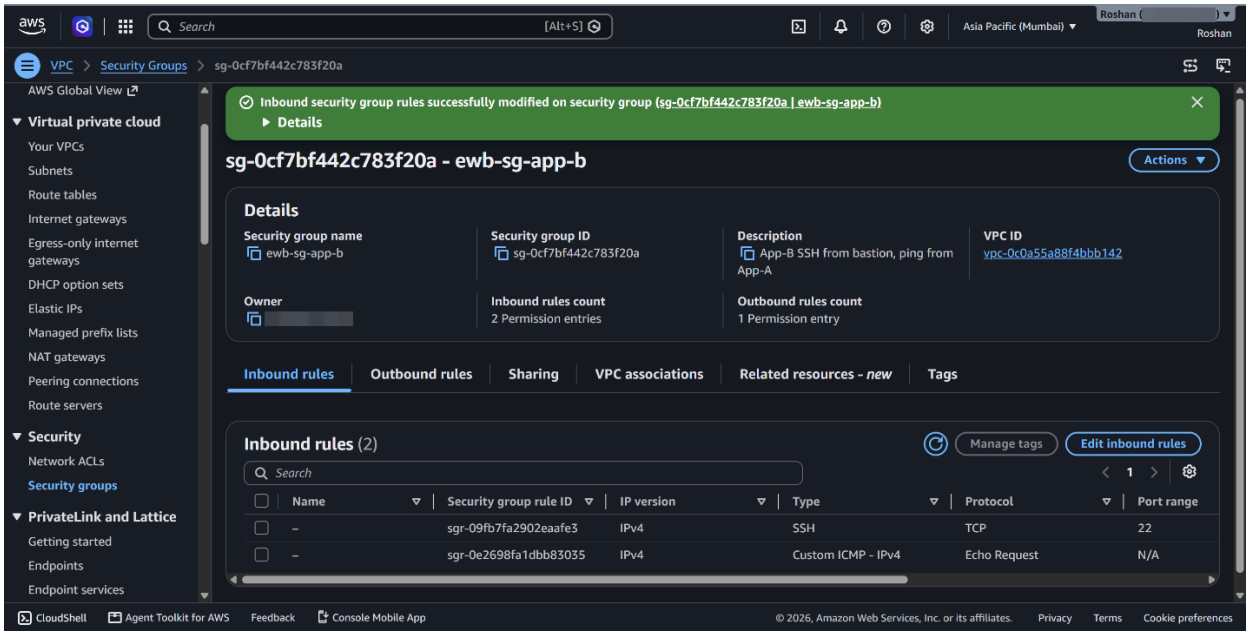


Figure 20. After adding the Echo Request rule back to ewb-sg-app-b, the group has two inbound rules again.

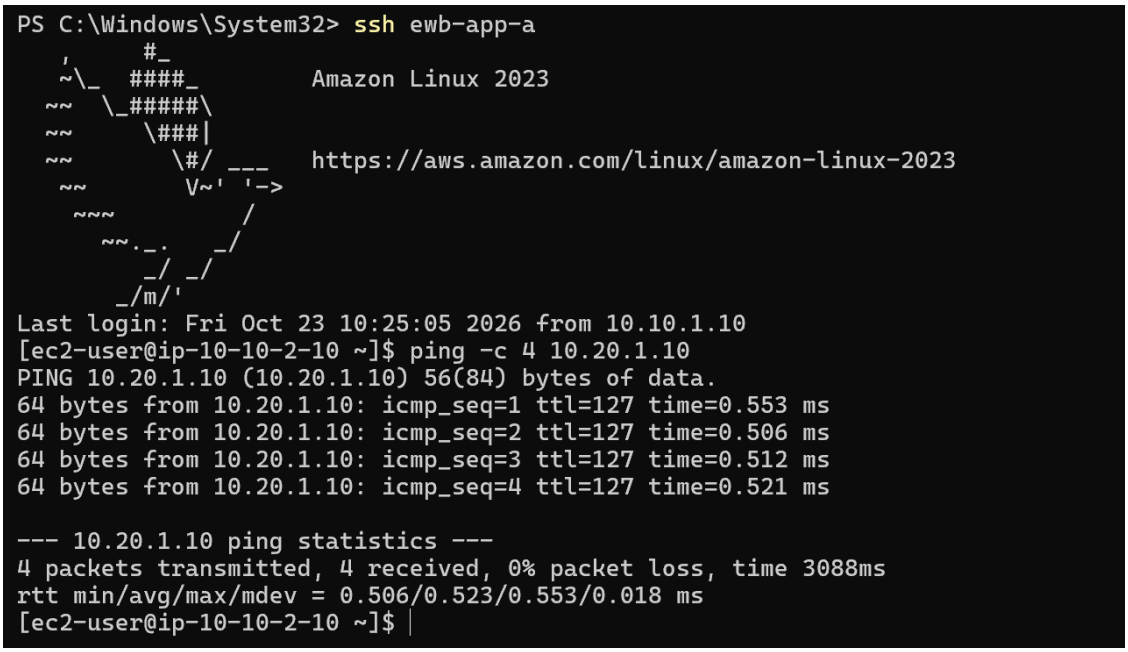


Figure 21. The same ping now gets 4 replies and 0% packet loss.

Even with peering and routes in place, the ping fails when the destination's security group blocks it. Security groups are stateful, so if a request is allowed, its reply is allowed automatically.

Forward path and return path

Forward path: App-A (10.10.2.10) pings 10.20.1.10. The route table ewb-rt-a-private sends the packet to the peering connection, which delivers it to VPC B. ewb-sg-app-b allows it.

Return path: App-B replies to 10.10.2.10. The route table ewb-rt-b-private sends the reply back through the peering connection. Without this return route, the reply would be lost and the ping would fail.